O-I-535b
SEPTEMBER 4, 1957
SUPERSEDING
Int. Fed. Spec. O–I–00535a (GSA–FSS)
April 9, 1956, and
Fed. Spec. O–I–535
January 27, 1953

FEDERAL SPECIFICATION

INSECTICIDE, LINDANE, WATER-DISPERSIBLE POWDER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 The insecticide covered by this specification is a 75-percent lindane water-dispersible powder designed to provide a water dispersion suitable for use as a residual spray for the control of mosquitoes, flies, sand flies, mites (chiggers), ticks, and fleas.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.**—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

RR–S–366—Sieves, Standard For Testing Purposes.

PPP–C–96—Cans, Metal, 28 Gage and Lighter.

PPP–D–760 — Drums and Pails, Metal (5 and 16.64 Gallon).

Federal Standards:

Fed. Std. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–C–124 — Containers (Cans, Pails, and Drums) Metal (For Other than Subsistence Items).

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

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(Copies of the Military specifications and standards required by contractors in connection with specific procurement functions, should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** — The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Federal Insecticide, Fungicide and Rodenticide Act.

(Copies may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Composition.—The insecticide shall be composed of lindane and inert ingredients, as follows:

3.1.1 *Lindane.* — The insecticide shall be compounded to contain a concentration of 75.0 percent lindane by weight, and the finished insecticide shall contain 548 mg. of organically bound chlorine per gram when tested as specified in 4.4.5. The lindane shall be the gamma isomer of 1,2,3,4,5,6-hexachlorocyclohexane of a purity not less than 99 percent. It shall be a white crystalline substance having a melting point of 112° to 113°C., when determined as specified in 4.4.1. The bidder shall stipulate, in a letter accompanying his bid, that the lindane used will meet the requirements of this paragraph.

3.1.2 *Inert ingredients.*—The inert ingredients shall be such biologically inert diluting, modifying, and conditioning agents, free of organically bound chlorine, as are needed to meet the requirements of this specification.

3.2 Form and color.—The insecticide shall be a free flowing powder, readily wettable with water and shall be of a light color such as white, cream, or light gray.

3.3 Foaming.—In the preparation of test suspensions of the insecticide powder, any foam produced shall not have such copiousness, stability, or other properties as would prevent the completion of the tests specified in 4.4.

3.4 Particle size.

3.4.1 *Average diameter.* — The surface-mean particle diameter of the insecticide powder, when determined as specified in 4.4.2.1, shall be not greater than 5.0 microns.

3.4.2 *Maximum diameter (under simulated storage conditions conducive to caking).*— Not less than 98 percent of the insecticide powder shall pass through a 74-micron (No. 200) U. S. Standard sieve, and not less than 100.0 percent shall pass through a 1-mm. (No. 18) U. S. Standard sieve, both sieves conforming to Federal Specification RR–S–366, when tested as specified in 4.4.2.2.

3.5 Dispersibility.—Not less than 95 percent of the insecticide powder incorporated in a suspension shall pass through a 250-micron (No. 60) U. S. Standard sieve conforming to Federal Specification RR–S–366, when tested as specified in 4.4.3.

3.6 pH.—A suspension of the insecticide powder prepared and tested as specified in 4.4.4 shall have a pH value not lower than 5.0 and not higher than 9.0. If pH is above 8.0, not more than 10.0 ml. of half normal hydrochloric acid shall be required to neutralize the alkalinity of a 20.0 gm. sample when titrated as specified in 4.4.4.

3.7 Workmanship.—The product shall be clean and uniform, and free of any defects which might impair its utility.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling a lot shall consist of all insecticide

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manufactured as one batch, and offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.* — A random sample of filled containers shall be taken from each lot by the Government Inspector in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.1.3 *Sampling for tests.*—Two containers shall be selected at random for each inspection lot. From each of the 2 containers 2-pound specimens shall be taken and placed in separate clean dry, metal or glass containers, sealed, marked and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.2 Inspection.

4.2.1 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined by the Government Inspector for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL–STD–105, the lot represented by the sample shall be rejected.

4.3 Lot acceptance tests. — The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4. If either specimen fails in one or more of the tests the lot shall be rejected.

4.4 Test procedures.

4.4.1 *Melting point of lindane.*—The melting point shall be determined by the capillary tube method, using a Thiele-Dennis, or equivalent, type melting tube, heated at a rate not to exceed 0.5°C. per minute. The temperature at which the sample becomes completely clear shall be the melting point.

4.4.2 *Particle size.*

4.4.2.1 *Average diameter.* — The surface-mean particle diameter shall be determined by the air-permeation method. The apparatus used shall be a Government approved air-permeation apparatus (see 6.2). The solid density of the powder, knowledge of which is prerequisite to particle fineness measurement by the air-permeation procedure, may be determined by any standard method. A suitable method is the conventional method of liquid immersion in a specific-gravity bottle having a ground-in thermometer, with distilled water as the immersion liquid. In the use of this method, care must be taken to guard against significant loss of water-soluble material; the pycnometer (specific-gravity bottle) should have a nominal minimum capacity of 25 ml. and the weight of sample used for observation should be as great as can be handled conveniently in the pycnometer used.

4.4.2.2 *Maximum diameter.* — This test, which is primarily a sieve test so designed that essentially all the material in an acceptable sample will pass through the sieve, involves a pretreatment to simulate storage conditions conducive to caking. The complete test consists of three steps: first, the pretreatment; second, the preparation of a suspension of the treated sample in water; and third, the actual screening and determination of the percentage of residue. The complete procedure shall be followed with each of two sieves, a relatively fine sieve (74 micron) adopted as a check in general granulometric quality of the product, and a relatively coarse sieve (1000-micron = 1-mm.) included as a safeguard against lumpiness. The three steps shall be performed in the following manner:

4.4.2.2.1 *Pretreatment.*—Pour 20 grams of the powder into a 250-ml. Griffin low-form

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beaker and level off the top surface of the powder bed without compacting the powder. Place within the beaker, on top of the powder, a loosely fitting rigid disk so formed and weighted as to exert upon the powder bed a pressure of 25 gm./sq. cm. Keep the powder thus under pressure in an oven at 55°C. for 24 hours. Take the sample from the oven, remove the weighting assembly, allow the powder to come to room temperature, and proceed to 4.4.2.2.2.

4.4.2.2.2 *Preparation of suspension.*—Prepare a 4 percent (total original solid) suspension by shaking together in a closed jar the pretreated powder and 480 gm. of distilled water. The jar in which the ingredients are shaken to form the suspension shall be a screw-cap glass jar of one U. S. liquid quart capacity, clear and transparent, substantially cylindrical, approximately 16 cm. in inside height (vertical clearance), and without abrupt shoulders or other shape features that would interfere with efficient transfer of the suspension from the jar to another vessel. (A U. S. quart fruit jar of suitable type, shape and dimensions should satisfy these requirements). The manner and extent of shaking to form the suspension shall be mechanical rotation of the jar, end over end, at the rate of 60 r.p.m. for five minutes; the axis of rotation shall be midway between the top and bottom of the jar and perpendicular to its vertical axis. When the shaking has been completed, proceed immediately to 4.4.2.2.3.

4.4.2.2.3 *Screening.*—Transfer the suspension to the sieve, which in preparation for the test must have been freed of any film of grease or other water-repellent material. Wash the residue on the sieve with an oscillating moderately vigorous spray of water for 10 minutes, and determine weight of the final residue by appropriate means, drying at 55° ± 2°C. to constant weight.

4.4.3 *Dispersibility.*—The dispersibility of the powder in water shall be tested in the following manner. Pretreat 20 gm. of the powder according to the procedure in 4.4.2.2.1. In preparation for the remainder of the test, make sure that the wire cloth of the sieve to be used is free of any film of grease or other water-repellent material, and is thoroughly wetted with water on both sides and kept so wetted until the moment for use. Prepare a suspension of the treated powder in water according to the procedure in 4.4.2.2.2 with the substitution of hard water (defined in 6.3) for distilled water. When the suspension has been made, immediately pour it upon the 250-micron sieve, taking care that the material still remaining in the jar is kept well stirred or shaken up, but taking care not to agitate or wash the sieve or the powder caught upon it. Utilize as nearly as possible the whole area of the screen, in order to reduce to a minimum any tendency of the particles to pile up. Upon completion of the pouring determine weight of residue by appropriate means, such as transfer to a beaker and evaporation to apparent dryness on a steam bath, ending with drying at 55° ± 2°C. to constant weight.

4.4.4 *pH.*—Make a 4 percent suspension in distilled water by the procedure in 4.4.2.2.2, but with 20 gm. of untreated powder. Allow the suspension to stand for 1 hour after preparation, then determine the pH value at 25°C. by any standard means. If pH is above 8.0, titrate the entire suspension with half normal hydrochloric acid, using phenolphthalein as an indicator.

4.4.5 *Organically bound chlorine.*—Weigh accurately a quantity of sample containing about 1.00 gm. of lindane, transfer to a 250-ml. Pyrex volumetric ask, and add about 200 ml. of chlorine-and thiophene-free benzene. Heat the contents of the flask to a boil. Cool at once to room temperature, make to volume with benzene, allow to settle and transfer a 10-ml. aliquot to a 300-ml. standard tapered Erlenmeyer flask. Add 50 ml. of 99-percent isopropanol and 2.5 gm. of metallic sodium in the form of ribbon or cut in small pieces. Connect to a reflux condenser and boil gently for at least ½ hour. Shake the flask

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occasionally. Eliminate excess metallic sodium by cautiously adding 10 ml. of 50-percent isopropanol through the condenser at the rate of 1-2 drops per second. Boil for an additional 10 minutes, and then add 60 ml. of water. Add 5 ml. of 30-percent hydrogen peroxide, a few drops at a time, through the top of the condenser. Heat the mixture in the flask to boiling and boil for 15 minutes. Cool the flask, and transfer the contents to a 400-ml. beaker. Add 2-3 drops of phenolphthalein solution, neutralize by adding nitric acid (1–1), and then add 10 ml. in excess. Remove any benzene from the solution by passing a gentle stream of air over the beaker. Add drop-wise, with stirring, a measured excess (25 ml. approximately) of 0.1 *N* silver nitrate solution. Coagulate the precipitate by heating on a steam bath for approximately ½ hour. Cool to room temperature and filter through a No. 42 Whatman filter paper and wash thoroughly with distilled water, receiving the filtrate in a 500-ml. Erlenmeyer flask. Add 5 ml. of ferric ammonium sulfate indicator solution and titrate the excess $AgNO_3$ with 0.1 *N* KCNS solution. Compute the net number of ml. of 0.1 *N* $AgNO_3$ consumed by the sample and calculated as follows:

$$a = \frac{b \times 3.546 \times 25}{s}$$

where:

a = milligrams of chlorine per gram of insecticide, and

b = net number of milliliters of 0.1*N* $AgNO_3$ consumed by the sample, and

s = weight of sample of insecticide in grams.

A blank determination (without sample) shall be made following the exact procedure given above but limiting the 0.1*N* $AgNO_3$ solution to 5 ml. in order to obtain a chloride correction value for all reagents used.

5. PREPARATION FOR DELIVERY

The definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—The product shall be furnished in 3-pound cans and in 20 pound containers.

5.1.1.1 *Three-pound cans.* — Three-pound cans shall be round, multiple friction plug, conforming to type V, class 2, of Federal Specification PPP–C–96. Plan B coating shall be furnished.

5.1.1.2 *Twenty-pound containers.*—Twenty (20) pounds of the product shall be packaged in a five-gallon metal pail in conformance with type II, class 1, 2, or 3 of Federal Specification PPP–D–760. The cover shall have the appropriate stock number embossed or metal stamped on the top or upon a corrosion-resistant coated metal tag which shall be secured to a lug. Pails shall be galvanized The exterior coating color shall conform to color number 24087 of Federal Standard No. 595.

5.1.2 *Level C.*—Unless otherwise specified, commercial packages are acceptable.

5.2 Packing.

5.2.1 *Level A and Level B.*—Three-pound cans shall be closed and packed in accordance with requirements of Appendix I of Federal Specification PPP–C–96. Twenty-pound containers need not be overpacked.

5.2.2 *Level C.*—Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—Shipping containers shall be marked as specified in Federal Standard No. 123.

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5.3.2 *Military agencies.* — Shipping containers shall be marked as specified in Military Standard MIL–STD–129.

5.3.3 *Labeling.* — The label data shall be lithographed or stenciled on container body with a weather-resistant and noncorrosive ink. Where plain olive-drab drums are employed a combination consisting of a stenciled label and a paper label may be used. The paper label shall be printed on a label made of strong paper affixed to the drum body with a water insoluble, noncorrosive adhesive and coated overall with a weather-resistant lacquer or enamel. The name of the product shall be stenciled on contained body when paper labels are used. The product shall be registered and labeled in compliance with the Federal Insecticide, Fungicide and Rodenticide Act. Unless otherwise specified, the label data shall read as follows:

INSECTICIDE, LINDANE (WATER-DISPERSIBLE POWDER)

Net weight _____

INGREDIENTS

Active Ingredients:

	Percent
Gamma Isomer of Benzene Hexachloride from Lindane _____	75
Inert Ingredients _____	25
Total _____	100

Manufacturer's name and address

Date

Contract number

Batch number

CAUTION

Warning: Harmful Vapor and Dust! May be Absorbed through skin! May Produce Irritation of Skin and Eyes! Avoid prolonged breathing of vapor, dust or spray mist. Do not leave in eyes or on skin. Wash thoroughly after all contacts with this insecticidal material. Avoid contamination of feed and foodstuffs.

Wettable powder concentrate must be diluted with water before use as directed under "Mixing Instructions".

This suspension should be used only under supervision of trained personnel. It is not to be used on humans and animals. Protective clothing and equipment should be worn when the insecticide is handled or used. Outdoor spraying should be done only with the operator upwind in relation to the surface or equipment sprayed.

MIXING INSTRUCTIONS

To insure good performance:

1. Prepare the finished spray as short a time as possible before use.
2. Add the proper amount of wettable powder concentrate to water and stir vigorously before use.
3. Power spray equipment should be provided with agitator to keep the insecticide in suspension while applying.
4. In using hand air pressure sprayers, shake container at frequent intervals to maintain uniform concentration of insecticide in spray and to avoid settling of suspended material which may result in clogging of spray nozzles.
5. The wettable powder spray shall again be thoroughly agitated for at least one minute immediately before use.

INSTRUCTIONS FOR USE

RESIDUAL SPRAY IN BUILDINGS

Mosquitoes and flies and sand flies:

Add 1⅔ ounces of insecticide per gallon of water (about 5 ounces to 3 gallons) and mix thoroughly. This makes a spray of about 1 percent lindane. Apply as a residual spray, wetting surfaces thoroughly just short of runoff. Usually 1 gallon is required for each 750 square feet, resulting in an actual lindane deposit of about 50 mg. per square foot.

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OUTDOOR SPRAYS (GROUND EQUIPMENT)

Mosquitoes and flies and sand flies:

For temporary control of these insects around campsites and bivouac areas, add ⅔ ounce of the insecticide per gallon (3 ounces in 8 gallons) and mix thoroughly. This makes a spray containing about 0.4 percent lindane. Apply lightly to vegetation, tree trunks and other surfaces at the rate of 5 to 10 gallons per acre depending on the amount of vegetation and other surfaces to be treated.

Mites (Chiggers), Ticks, and Fleas:

Add ⅔ ounce of insecticide to 1 gallon of water as above. For mites (chiggers) and ticks apply to ground litter and low vegetation at the rate of 10 to 20 gallons per acre depending on the amount of vegetation and litter. For flea control, apply at the same rate to infested grounds and under buildings.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein (see 5.1, 5.2, and 5.3).

6.2 Air permeation apparatus. — An approved apparatus is the self-calculating air-permeation apparatus of the U. S. Department of Agriculture (Patent No. 2,261,802), or equal. The Sub-Sieve Sizer, manufactured by the Fisher Scientific Company, may be used as the test instrument. Each individual apparatus is subject to inspection by the purchasing agency as a check on calibration and performance.

6.3 Hard water.—For purposes of test, hard water is defined as water of the following composition, designed to provide a hardness of 342 p.p.m. calculated as calcium carbonate:

Calcium chloride, anhydrous	0.3037 gram
Magnesium chloride hexahydrate	0.1388 gram
Distilled water, to make	1 liter

6.4 Transportation description.—Transportation description applicable to these items is:

Insecticides, not otherwise indexed by name, other than agricultural.

Carload weight 36,000 pounds.

Truckload volume weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q M
Navy—Sh
Air Force.

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Copies of this specification may be purchased for each.

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